

POPULATION DYNAMICS IN CHEMOSTAT: A TWO SPECIES MODELING APPROACH

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Abstract

This project's mainstream is to construct a mathematical model for a Chemostat to describe the dynamics of two interacting microbial species through model analysis. The system's dynamics are described using a system of non-linear ordinary differential equations given by Odo Deikman[1], considering the growth dynamics of each species and their interaction with the substrate. The model incorporates the Monod growth kinetics[2] for each species and explores their dynamics under varying conditions. Specifically, the project focuses on analyzing the impact of substrate availability, dilution rate, and species interaction on the population dynamics within the Chemostat system. By incorporating interactions between two species, this study seeks to explore the behavior of microbial populations under different ecological conditions.

1 Introduction

Chemostat systems are widely used in microbiology and biotechnology to study microbial growth dynamics under controlled environmental conditions. A chemostat is an automated bioreactor that contains metabolic products, microorganisms, and leftover nutrients that are continuously removed while fresh medium is added at the same rate to keep the volume constant (Novick and Szilard, 1950). The chemostat provides a powerful means of systematically investigating how growth rate impacts processes of the cells such as gene expression and metabolism and the regulatory networks that control the rate of cell growth.

Moreover, cells grown in chemostats for generations can be used to study their adaptive evolution in environmental conditions that limit cell growth (Ziv et al.) Understanding the dynamics of microbial populations in chemostats is crucial for various applications, including wastewater treatment, bioremediation, and bioprocess optimization. The chemostat serves as a controlled environment for studying microbial growth and competition. Through mathematical analysis and computational simulations, I aim to investigate stability, coexistence,

extinction, and competitive outcomes within the two-species Chemostat system by visualizing different conditions from the table belongs to the paper by Butler and G. S. K. Wolkowicz[5].

1.1 Motivation

For decades, researchers have been interested in the growth of bacterial and cell populations due to their importance in industry and ecology and the complexity of their dynamics. Understanding the coexistence of different populations is key in these studies. In a 1979 paper, Stephanopoulos et al. analyzed the dynamics of two microbial populations competing for a single substrate. Before their work, Aris and Humphrey (1977) showed that two populations could coexist when the substrate concentration is maintained at the break-even point for both and the dilution rate remains constant. Hsu (1978) extended this result for multiple populations using Lyapunov functions.

However, these scenarios present a delicate balance where even a small deviation in the dilution rate can lead to the extinction of one population (El Hajji and Rapaport, 2009). Stephanopoulos et al. found that this extinction can occur due to fluctuations in controlling the dilution rate in chemostats. Depending on the dilution rate and initial conditions, either population may become extinct.

The dilution rate is a critical factor, and when it is lower than a microorganism's growth rate, that microorganism will thrive. The Competitive Exclusion Principle (CEP) is a key concept in this context. It suggests that in a system with multiple populations competing for a single substrate, only one population will survive. The one with the lowest break-even concentration, whose growth rate matches the dilution rate, will outcompete the others. Many papers have focused on proving the CEP under various assumptions about growth functions and removal rates. Here, my objective is to explain their thought graphically to understand the population growth over time while there is a change in the values of the parameters.

2 Model Development

The dynamics of the chemostat system are described by a system of ordinary differential equations (ODEs), which represent the changes in the concentrations of the two microbial species (X and Y) and the substrate (S) over time. The model equations are as follows[1]:

$$\frac{dX}{dt} = (\eta_{\text{res}}\beta(S) - D)X \quad (1)$$

$$\frac{dY}{dt} = (\eta_{\text{inv}}\beta(S) - D)Y \quad (2)$$

$$\frac{dS}{dt} = D(S_0 - S) - \beta(S)X - \beta(S)Y \quad (3)$$

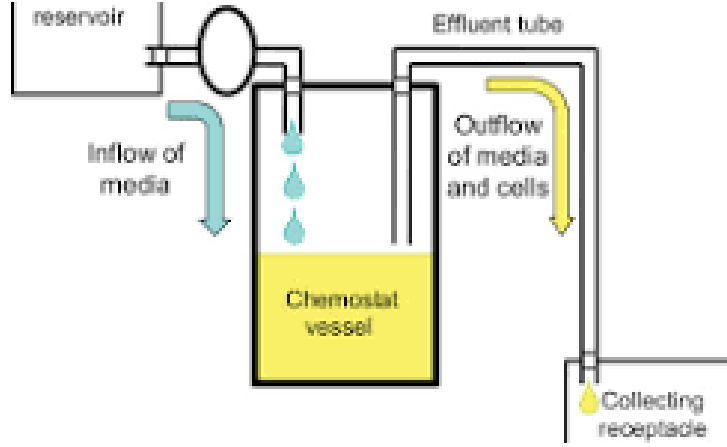


Figure 1: Chemostat

Where:

- X and Y represent the concentrations of species 1 and species 2, respectively.
- S represents the concentration of the substrate.
- $\beta(S)$ is the growth function of the microbial species, defined as:

$$\beta(S) = \frac{aS}{1 + ahS} \quad (4)$$

where a and h are parameters governing the growth kinetics. This function describes how the growth rate of each species depends on the substrate concentration (S). Parameters a and h determine the functional form of the relationship.

- η_{res} and η_{inv} are the conversion efficiency of species X and Y , respectively. Which represents the efficiency of converting the substrate into biomass.
- D is the dilution rate of the chemostat. If the dilution rate and in the case where it is smaller than the growth rate of the micro-organism that microorganism will grow.
- S_0 is the concentration of the substrate in the feed bottle.
- The First equation represents the change in the population X , the resident species (X) over time. The growth rate of X depends on the difference between its maximum growth rate ($\eta_{\text{res}}\beta(S)$) and the dilution rate (D). If $\eta_{\text{res}}\beta(S) > D$, X will increase. Otherwise, it will decrease.[1]

- The Second equation describes the change in the population density of the invading species (Y) over time. The equation follows the same logic as the previous one, with the growth rate determined by the difference between its maximum growth rate ($\eta_{\text{inv}}\beta(S)$) and the dilution rate (D).
- The Third equation models the change in substrate concentration (S) over time. The change is influenced by the input of new substrate ($D(S_0 - S)$), and the consumption of substrate by the resident and invading species ($\beta(S)X$ and $\beta(S)Y$, respectively).

2.1 Assumption

1. S is non-reproducing.
2. Input concentration and dilution rate is constant
3. There is perfect mixing in the vessels.
4. The death rate of any species is insignificant compared to D.

2.2 Model Interaction

The equations are **interdependent**: changes in one variable can affect the others. For example, as the resident(X) and invading species(Y) consume substrate(S), the available substrate decreases over time, impacting the growth rates of both species. Likewise, changes in substrate concentration affect the dynamics of species populations. The interaction is influenced by the value of the parameters and the dynamics changes with them. For the above model, three possible scenarios can happen,

1. Coexistence: When both species have non-zero populations at steady state being in a stable system.
2. Invasion: When any of the population dominates another and causes extinction of it. It depends on the value of conversion efficiency, growth function, and dilution rate. Invasion happens, When one population outcompetes another population leading to dominance or exclusion of the resident, if the invading species has a higher conversion efficiency than the resident species, if the invading species has a lower break-even concentration compared to the resident species and if invading species starts with higher initial population.
3. Extinction: When one or both species die out due to insufficient resources or unfavorable environment(ex-Dilution rate).

Two crucial parameters for this model are Dilution Rate, D, and Conversion efficiency, η . Giving a short overview of them,

2.2.1 Role of Conversion Efficiency

- **Higher Conversion Efficiency:** When conversion efficiency is high, a larger proportion of the available substrate is converted into biomass, leading to higher population densities. In a chemostat, higher conversion efficiency allows microbial populations to thrive even when substrate concentrations are low, as they can efficiently utilize the available resources for growth.
- **Lower Conversion Efficiency:** Conversely, lower conversion efficiency results in less efficient utilization of resources and lower biomass production per unit substrate consumed. In a chemostat, lower conversion efficiency may limit the population growth rate, particularly when substrate concentrations are low, leading to lower population densities.
- **Effects of Changes in Conversion Efficiency:** Changes in conversion efficiency can significantly affect population dynamics in a chemostat system. Increasing conversion efficiency generally leads to higher population densities, while decreasing conversion efficiency may result in lower population densities. Changes in conversion efficiency can also affect the competition dynamics between different species in the chemostat. A species with higher conversion efficiency may outcompete others for limited resources, leading to shifts in species composition and population structure. And, by changing the value of it, for several cases, I have observed an increase, decrease, and even extinction of the species.

2.2.2 The role of Dilution rate, D

The dilution rate in a chemostat system refers to the rate at which the culture medium (containing both microbial populations and nutrients) is removed and replaced with fresh medium per unit of time. It's typically denoted by the symbol D and is expressed in units of reciprocal time (e.g., per hour or day)[3]. The impacts of it are given below,

- **Population Density:** When the dilution rate is too high microbial populations from the system get removed at a higher rate, which results in lower population densities over time. Conversely, lower dilution rates initiate populations to grow to higher densities.
- **Nutrient Availability:** The dilution rate also affects the substrate concentration or nutrients in the system. Higher dilution rates result in faster depletion of nutrients or substrate due to their rapid removal, leading to a decrease in the substrate concentrations that are available for microbial growth.
- **Stability of Steady States:** The dilution rate can influence the stability of steady states in the system. Changes in dilution rate can lead to shifts in the balance between growth and removal of microbial populations, potentially destabilizing steady states and altering system behavior.

- **Competition and Coexistence:** Dilution rates can impact the outcomes of competition between microbial species. Higher dilution rates may favor fast-growing species that can outcompete others for limited resources, leading to the exclusion of less competitive species. Conversely, lower dilution rates may promote coexistence by allowing populations to reach equilibrium.

Overall, the dilution rate plays a fundamental role in shaping the dynamics and behavior of microbial populations in chemostat systems. Understanding its impact is essential for designing, operating, and analyzing continuous culture systems in various fields of research and application

3 RESULTS

3.1 Stability Analysis

The model has several steady states that represent equilibrium points where species' populations and substrate concentration remain constant over time. Now, I will determine the conditions for coexistence or invasion. Setting the rate of change of X , Y , and S of equation(1,2,3) is equal to zero we get, The model will be solved numerically using appropriate computational techniques, such as ordinary differential equation solvers. The dynamics of the system will be analyzed by varying parameters such as the dilution rate, initial substrate concentration, and the parameters a and h in the growth function. Sensitivity analysis will be performed to identify key factors influencing the stability and behavior of the system. In the context of this project on a chemostat model for two species and the analysis of conditions for coexistence or invasion, the mathematical model (section 2) and the results (section 4) are crucial components. Let's delve deeper into these sections:

$$\eta_{\text{res}}X\beta(S) - DX = 0 \quad (5)$$

$$\eta_{\text{inv}}Y\beta(S) - DY = 0 \quad (6)$$

$$D(S_0 - S) - \beta(S)X - \beta(S)Y = 0 \quad (7)$$

This set of equations can be simplified to find potential equilibrium points for the system. For each equation, consider the possibilities:[4].

- $X(\eta_{\text{res}}\beta(S) - D) = 0$. This equation is true if $X = 0$ or $\eta_{\text{res}}\beta(S) = D$.
- $Y(\eta_{\text{res}}\beta(S) - D) = 0$. This equation is true if $Y = 0$ or $\eta_{\text{inv}}\beta(S) = D$.
- $D(S_0 - S) - \beta(S)X - \beta(S)Y = 0$. This equation is an expression of substrate balance.

$$\text{equilibrium_point1} := \begin{cases} X = 0, \\ Y = 0, \\ S = S_0 \end{cases}$$

$$\text{equilibrium_point2} := \begin{cases} S = -\frac{D}{a(Dh-\eta_{\text{res}})}, \\ X = \frac{\eta_{\text{res}}(DS_0ha-S_0a\eta_{\text{res}}+D)}{a(Dh-\eta_{\text{res}})}, \\ Y = 0 \end{cases}$$

$$\text{equilibrium_point3} := \begin{cases} S = -\frac{D}{a(Dh-\eta_{\text{inv}})}, \\ X = 0, \\ Y = \frac{\eta_{\text{inv}}(DS_0ha-S_0a\eta_{\text{inv}}+D)}{a(Dh-\eta_{\text{inv}})} \end{cases}$$

3.2 Finding Eigenvalues

Now, first expressing the above equation in the form of a Jacobian Matrix and by solving linearly, I find the eigenvalues of each equilibrium point. The Jacobian

$$\text{Matrix: } \mathbf{J} = \begin{bmatrix} \eta_{\text{res}}X\beta(S) - D & 0 & \frac{\eta_{\text{res}}aX}{Sah+1} - \frac{\eta_{\text{res}}XhSa^2}{(Sah+1)^2} \\ 0 & \eta_{\text{inv}}\beta(S) - D & \frac{\eta_{\text{inv}}aY}{Sah+1} - \frac{\eta_{\text{inv}}a^2YSh}{(Sah+1)^2} \\ \frac{-Sa}{(Sah+1)} & \frac{-Sa}{(Sah+1)} & -D - \frac{aX}{(Sah+1)} + \frac{a^2SXh}{(Sah+1)^2} - \frac{aY}{(Sah+1)} + \frac{a^2SYh}{(Sah+1)^2} \end{bmatrix}$$

$$\text{E1} = \begin{bmatrix} -\frac{DahS_0-aS_0\eta_{\text{inv}}+D}{ahS_0+1} \\ -D \\ -\frac{DahS_0-aS_0\eta_{\text{res}}+D}{ahS_0+1} \end{bmatrix}$$

$$\text{E2} = \begin{bmatrix} \frac{(\eta_{\text{inv}}-\eta_{\text{res}})D}{\eta_{\text{res}}} \\ -D \\ -\frac{D^2h^2aS_0+2\eta_{\text{res}}DahS_0-aS_0\eta_{\text{res}}^2+D^2h-D\eta_{\text{res}}}{\eta_{\text{res}}} \end{bmatrix}$$

$$\text{E3} = \begin{bmatrix} -\frac{(\eta_{\text{inv}}-\eta_{\text{res}})D}{\eta_{\text{inv}}} \\ -\frac{D^2ah^2S_0-2DahS_0\eta_{\text{inv}}+aS_0\eta_{\text{inv}}^2+D^2h-D\eta_{\text{inv}}}{\eta_{\text{inv}}} \\ -D \end{bmatrix}$$

3.2.1 Explanation

Since all the parameters are positive then the equilibrium points are also positive but their values depend on the parameters. According to the eigenvalues above the three scenarios depend on the stability of the eigenvalues. This proves the postulates from [7]- A system has at most three equilibrium points on the boundary of the positive quadrant. The trivial equilibrium point $E(0,0,S_0)$ always exists and it is a locally asymptotically stable 'microorganism-extinction' point when $\beta(S_0)-D < 0$ and the other two equilibrium points of above form when $\beta(S_0)-D > 0$.

Now, from the eigenvalues we can have the following interpretation:

- **Stable Steady State:** If all eigenvalues of the Jacobian matrix evaluated at the steady state have negative real parts, the steady state is stable[6]. This means that small perturbations around the steady state will decay over time, and the system will return to the steady state. **In a stable system, the coexistence of the two species is possible.** Mostly the

eigenvalues above have negative real parts, so it is obvious to assume they will be stable but it solely depends and varies with parameters. Initially, The stable steady state allows both species to persist in the environment without being displaced by small fluctuations or perturbations. Coexistence occurs when the growth rates of both species are balanced, and no species has a competitive advantage over the other under the prevailing environmental conditions. For a steady and stable state, both populations will be growing simultaneously. In my simulation sector, I have shown the scenario of coexistence for different values of parameters.

- **Unstable Steady State:** The steady state is unstable if any eigenvalue of the Jacobian matrix has a positive real part[6]. This indicates that small perturbations around the steady state will grow over time, leading the system away from the steady state. **In an unstable system, one species may invade the environment and outcompete the other.** The unstable steady state acts as a barrier to coexistence, as any deviation from equilibrium leads to displacing one species by the other. Invasion occurs when one species has a competitive advantage over the other due to a higher growth rate or a lower sensitivity to environmental conditions. For different values of parameters from the above value of eigenvalues, it can easily be determined what is happening in the unstable state.

3.3 Numerical Simulation

Now, I will do the numerical stimulation to visualize the behavior of the chemostat model under various conditions. Here, for different values of parameters, I will demonstrate the population dynamics. To solve this, I use Maple and R software and to visualize them I have used the values of parameters from the following table.

D	a	h	S_0	η_{res}	η_{inv}	X_0	Y_0	S0
0.25	0.1	0.05	10	0.7	0.8	1	.5	5
0.5	0.01	0.05	25	0.9	0.8	1	.5	5
0.05	0.01	0.05	25	0.8	0.8	2	1	5

Table 1: Parameters

4 DISCUSSION

Before proving the conditions given by Butler and G. S. K. WOLKOWICZ , it is a must to understand the break-even concentration of a substrate.

4.1 Break-Even Concentration

Break-even concentration is a substrate concentration where the growth function of any microbial population in a chemostat becomes zero. It is also known

as critical substrate concentration. In other words, it is the minimum substrate concentration required for a population to sustain a steady state without growth or decline. To find the break-even concentration, from equation 5,6,7 The breakeven concentration is of the form(when $X \neq 0$ and $Y \neq 0$) we get,

$$\eta_{\text{res}}\beta_S = D \quad (8)$$

$$\eta_{\text{res}}\beta_S = D \quad (9)$$

Solving the above two equations we get the break-even concentration for X and Y as follows,

$$S_{\text{res}} = \frac{D}{\eta_{\text{res}}a - Dah} \quad (10)$$

$$S_{\text{inv}} = \frac{D}{\eta_{\text{inv}}a - Dah}. \quad (11)$$

Now, based on the break-even substrate, I will try to explain the three scenarios by explaining the conditions stated by Butler and G. S. K. Wolkowicz[5].

1. **The breakeven concentration of S, $S_b < S_0$, population coexists [5]:** When both populations express stable populations, then coexistence happens. And their break-even concentrations are close to each other and also both less than the initial substrate concentration S_0 .

From the table(1st row), by calculating we get,

$$S_{\text{res}} = 3.588 < S_0 = 10 \quad (12)$$

$$S_{\text{inv}} = 3.13 < S_0 = 10 \quad (13)$$

i.e Then both populations will coexist. Since $S_{\text{res}} > S_{\text{inv}}$, so over time X population will die out and will be invaded by Y [5]. **[Fig-2]** Opposite happens when $S_{\text{res}} < S_{\text{inv}}$ **[Fig-3]**

2. **When, the breakeven concentration of $S_b > S_0$, population dies out [5]:** Invasion and extinction both or any could happen due to the influence of parameters like if the species will invade other species if it has a higher conversion efficiency than the other or if one species has a lower break-even concentration compared to other species and if invading species starts with higher initial population.

i.e if $\eta_{\text{res}}\beta(S) > D$, **X will increase** or $\eta_{\text{inv}}\beta(S) > D$, **Y will increase**.

Extinction occurs when one or both species are unable to maintain their population densities, eventually leading to their demise.

This can happen when: If the dilution rate is too high relative to the species' growth rates $\eta_{\text{res}}\beta(S) < D$ or $\eta_{\text{inv}}\beta(S) < D$, the populations may be washed out of the system. Suppose the initial substrate concentration or the substrate supply is too low. In that case, neither species may

be able to maintain a viable population, if the growth function ($\beta(S)$) is very limited by the available substrate concentration. In some cases, one species may exclude the other so completely that both eventually go extinct if the winning species' break-even concentration is greater than the initial substrate concentration.

From the table, we get(2nd row),

$$S_{\text{res}} = 57.14 > S_0 = 25 \quad (14)$$

$$S_{\text{inv}} = 64.51 < S_0 = 25 \quad (15)$$

i.e the substrate is insufficient to support either species at their break-even concentrations, also $\eta_{\text{res}}\beta(S)=0.499 < D=.5$ and $\eta_{\text{inv}}\beta(S)=0.493 < D=.5$ leading to extinction and both populations will die out[5].**[Fig-4]**

3. When $S_{\text{res}} > S_{\text{inv}}$ or $S_{\text{res}} < S_{\text{inv}}$ both population coexists at a positive equilibrium. From the 3rd row of the table, I configured the plot**[Fig-5]**. Where, over time the X,Y, and S maintain a stable state.

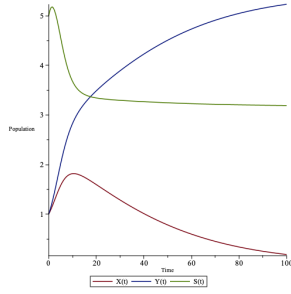


Figure 2: Y is dominant

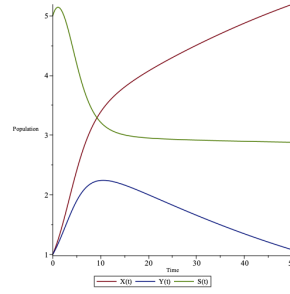


Figure 3: X is dominant

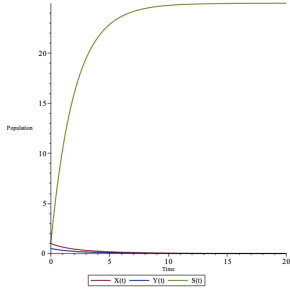


Figure 4: X and Y are dying out

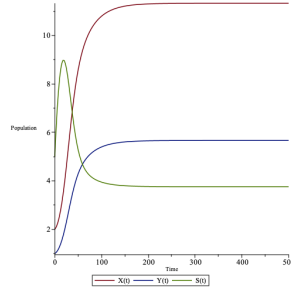


Figure 5: Stable Coexistence

4. When S_0 is extremely high and the conversion efficiencies > 1 at both populations tend to increase but reach stability over time **[Fig-6, Fig-7]**. But when $\eta_{\text{res}} > \eta_{\text{inv}}$ then the Y population shows an increase at first but slowly it begins to decrease and eventually dies out and the same happens with X population when $\eta_{\text{res}} < \eta_{\text{inv}}$.**[Fig-8, Fig-9]**

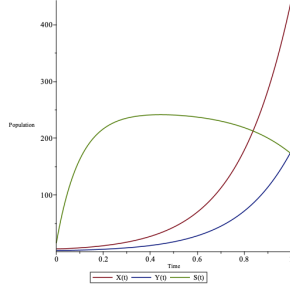


Figure 6: when S_0 and both $\eta > 1$

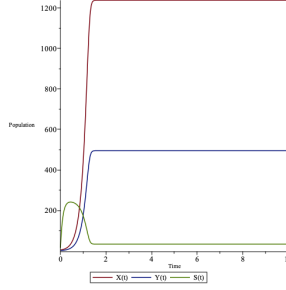


Figure 7: Stable Coexistence

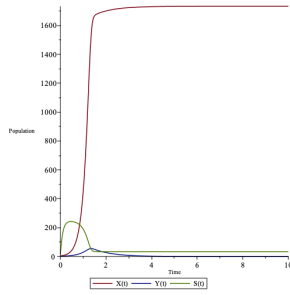


Figure 8: $\eta_{\text{res}} > \eta_{\text{inv}}$

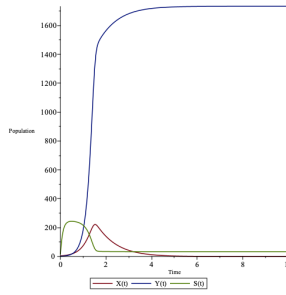


Figure 9: $\eta_{\text{res}} < \eta_{\text{inv}}$

5 Conclusion

In the model, I demonstrated how the changes in parameters influence the nature of populations X and Y in the presence of substrate S. When the value of conversion efficiency, dilution rate, and initial concentration differs the population of X and Y show different scenarios and satisfies the conditions of stability [6] and simulations [5]. Furthermore, my motivation was to justify the conditions of the break-even concentration of X and Y to find stability or extinction. In condition-1,2,3, I tried to demonstrate how important is the break-even concentration, below that the population can't be sustained and it decreases whereas above it the population can grow and reach a steady state where the growth rate is equal to the dilution rate. So, to maintain the population stable, keeping the concentration of S in the right amount is a must otherwise drastic changes or extinction may happen. By developing and analyzing the proposed model, this project aims to provide insights into the dynamics of two-species Chemostat systems and their response to environmental changes. Three scenarios coexistence, invasion, and extinction solely depend on the simulation above. I focused on demonstrating the interaction between species X and Y to understand and visualize which condition may drive the coexistence, invasion, or extinction. Understanding the interactions between microbial species and their environment is crucial for various applications in biotechnology and environmental engineering. The findings of this study may contribute to the optimization of Chemostat op-

eration and the design of microbial processes for industrial and environmental applications. The analysis of the model focuses on steady-state solutions and the conditions for stable coexistence or invasion. By examining the model parameters and their effects on the system's behavior, I identified critical factors that drive the competition between the species and understand the ecological implications of the model.

6 References

- [1] Odo Diekmann, Mathematical Modeling of Population Dynamics, Branch Center publication, Volume 63, Institute of Mathematics Polish Academy of Science, Warszawa 2004
- [2] J. Monod, La technique de culture continue theorie et applications, Ann. Inst. Pasteur 79 (1950) 390–410.
- [3] S.B. Hsu, S. Hubbell, P. Waltman, A mathematical theory for single-nutrient competition in continuous cultures of microorganisms, SIAM J. Appl. Math. 32 (1977) 366–383.
- [4] J.A.J Metz and O.Diekmann, The Dynamics of Physiologically Structured Populations, Lectures Notes in Biomathematics 68.Springer, Berlin,1986.
- [5] G. J. BUTLER AND G. S. K. WOLKOWICZ, Exploitative Competition in a Chemostat for Two Complementary, and Possibly Inhibitory, Resources.
- [6] Julien Arino, Sergei S. Pilyugin and Gail S. K. Wolkowicz, CONSIDERATIONS ON YIELD, NUTRIENT UPTAKE, CELLULAR GROWTH, AND COMPETITION IN CHEMOSTAT MODELS, Canadian Applied Mathematics Quarterly Volume 11, Number 2, Summer 2003.
- [7] Mengnan Chi1 and Wencai Zhao,Dynamical Analysis of Two Microorganism and single Nutrient Stochastic Chemostat Model with Monod-Haldane Response Function,Hindwai Complexity,Volume 2019,Article ID 8719067.